# SENTIMA RISK ENGINE

**A TACTICAL, HIGH-FREQUENCY STOCK INDEX REBALANCER**

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# PROBLEM STATEMENT

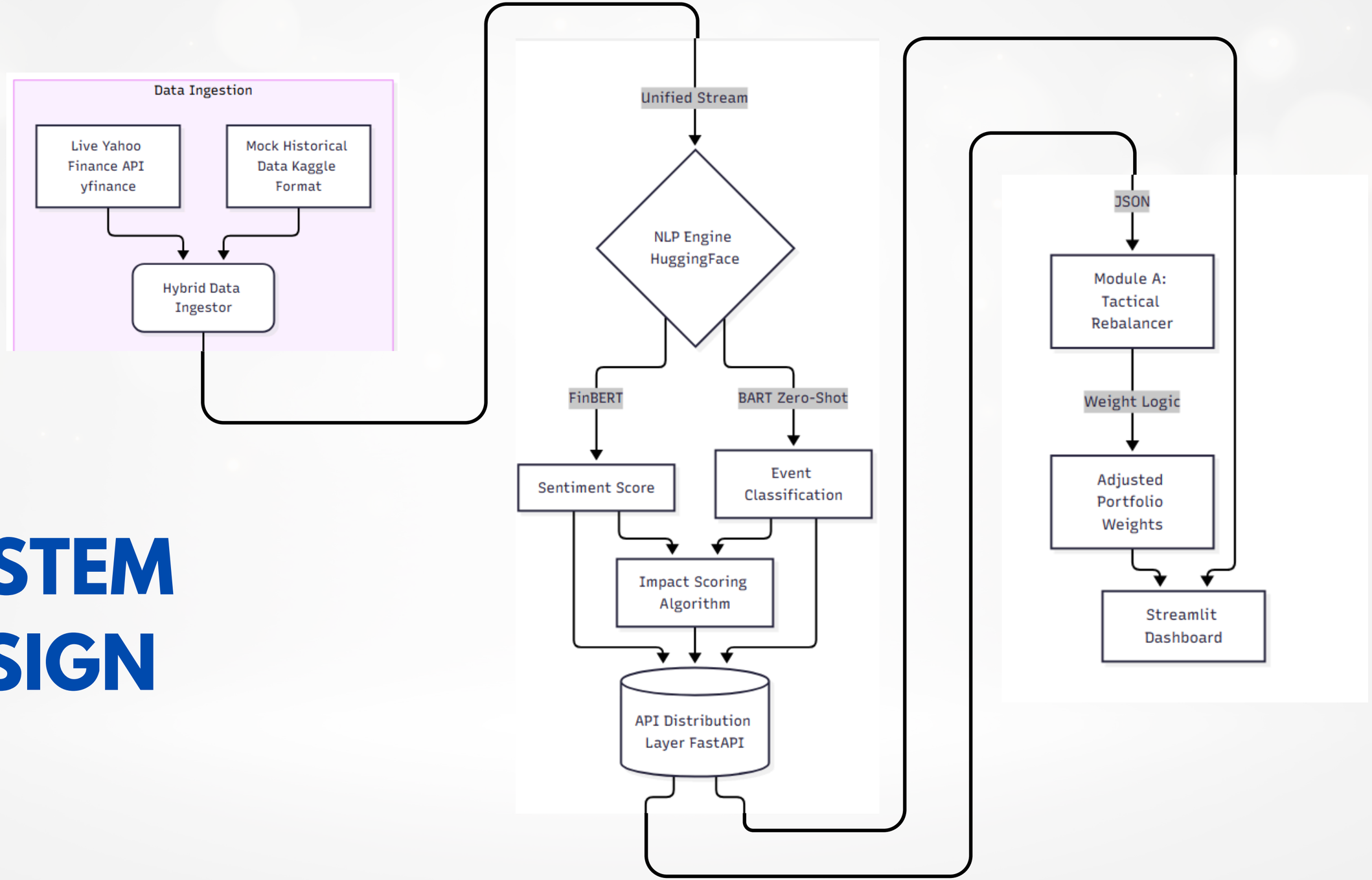
The Business Problem:

- Information Overload & Latency: Financial markets react instantly to global events, but manually analyzing the massive volume of unstructured data (news, social media) is slow and inefficient.
- Lack of Structured Signals: Raw text data lacks the structure needed for automated, quantitative decision-making.
- The Gap: There is a critical need for a system that can continuously ingest this qualitative noise and instantly translate it into quantified risk metrics for immediate portfolio action.

# SOLUTION APPROACH

- Unified AI/NLP Risk Engine:  I am Developing a central, automated pipeline that ingests real-time text from diverse sources (e.g., financial news, social feeds).
- Automated Signal Generation: Utilizing Natural Language Processing (NLP), the engine analyzes the text to output three structured, machine-readable metrics:
    - Sentiment Score (Positive/Negative/Neutral)
    - Event Classification (e.g., Geopolitical, Macroeconomic)
    - Impact Score (Predicted market severity, 1-10)
- Actionable Downstream Modules: To demonstrate real-world value, the structured signals feed directly into specialized modules:
    - Tactical Index Rebalancing: Dynamically adjusting portfolio weights of a mock stock index in real-time based on fluctuating sentiment.

# SYSTEM DESIGN



# IMPLEMENTATION HIGHLIGHTS

| KEY MODULES | TECH CHOICES | WHY? |
| --- | --- | --- |
| • DATA INGESTION MODULE | • HUGGINGFACE AI MODELS (FINBERT & BART) | • HUGGINGFACE (FINBERT & BART): FINBERT ACCURATELY PARSES COMPLEX FINANCIAL JARGON FOR SENTIMENT, WHILE BART ENABLES ZERO-SHOT EVENT CLASSIFICATION WITHOUT NEEDING CUSTOM MODEL TRAINING. |
| • AI/NLP RISK ENGINE (THE CORE) | • FASTAPI & UVICORN | • FASTAPI: PROVIDES A LIGHTNING-FAST, ASYNCHRONOUS REST API LAYER TO SMOOTHLY DECOUPLE THE HEAVY AI ENGINE FROM THE DOWNSTREAM APPLICATIONS. |
| • API DISTRIBUTION LAYER | • STREAMLIT | • STREAMLIT: ENABLES RAPID DEVELOPMENT OF THE INTERACTIVE DATA DASHBOARD AND LIVE DEMONSTRATION USING PURE PYTHON. |
| • TACTICAL INDEX REBALANCER (MODULE A) | • PANDAS & NUMPY | • PANDAS & NUMPY: DELIVERS THE HIGH-SPEED DATA MANIPULATION NECESSARY FOR EXECUTING THE REAL-TIME TACTICAL REBALANCING ALGORITHMS. |
| • INTERACTIVE DASHBOARD | • YFINANCE | • YFINANCE: ENABLES FETCHING REAL-TIME MARKET DATA TO EXECUTE OUR TACTICAL REBALANCING LOGIC AGAINST LIVE STOCK PRICES (WITH A MOCK DATA FAIL-SAFE MECHANISM) |

# KEY RESULTS







## OUTPUTS & METRICS

- Real-time Signal Translation: Successfully translates unstructured qualitative text into actionable, quantitative JSON signals in under 1.5 Sec
- Automated Risk Mitigation: Tactical Rebalancer successfully triggers automated capital protection routines when high-severity events (Impact Score ≥ 7) are detected.
- Model Performance: Our pipeline leverages FinBERT to achieve ~88% accuracy in finance-specific sentiment classification, maintaining an ultra-low inference latency of <1.5 seconds per batch and a throughput capacity of 50+ headlines per second.

## EFFICIENCY GAINS VS. NAIVE APPROACH

- Naive Approach (Traditional): Human analysts manually monitoring news feeds, resulting in subjective risk analysis, high latency, and delayed manual portfolio adjustments.
- Our AI Approach:
  1. Speed: Instant, automated ingestion and rebalancing execution.
  2. Accuracy: Objective, un-biased sentiment scoring tailored to financial jargon (via FinBERT) rather than generic human interpretation.
  3. Scale: Can process thousands of articles a minute, whereas a human can process maybe one.

# DOMAIN IMPACT

**From a Business Standpoint:**

**By instantly converting unstructured global news into quantifiable risk signals, our engine allows financial institutions to execute high-speed defensive trades before a human analyst even reads the headline—shifting risk management from reactionary to proactive and protecting vital capital during market shocks.**

# WHY IT MATTERS ?

1. **Capital Protection & Alpha Generation:** Instantly preserves capital during market shocks and captures upside during bullish shifts through automated, real-time portfolio adjustments.
2. **Eliminating Human Latency:** Replaces slow, subjective human analysis with millisecond-speed automated execution, beating the broader market to breaking news.
3. **Quantifying the Qualitative:** Solves a major Wall Street challenge by transforming messy, unstructured text into clean, machine-readable risk metrics for algorithmic trading.
4. **Enterprise Scalability:** The decoupled API architecture allows seamless integration into any institution's existing trading algorithms, compliance monitors, or retail apps.

# LIMITATIONS & ASSUMPTIONS

- Data Constraints:  While we successfully integrated live market data via yfinance, we assumed simulated news inputs for the NLP side for this hackathon.

- Zero-Shot Limitations:  While BART provides excellent zero-shot classification, it lacks the deep, nuanced accuracy of an event classifier fine-tuned exclusively on decades of historical market data.

- Simplified Rebalancing:  Our mock index logic assumes zero transaction fees and infinite liquidity, which are not reflective of real-world trading constraints.

# NEXT STEPS & FUTURE WORK

- Live WebSocket Integration: Replace static polling with live streaming data ingestion (e.g., direct Twitter/X or news provider WebSockets) for true millisecond-level reaction times.

- Model Fine-Tuning: Train a custom, lightweight LLM specifically on historical Impact Scores to better predict exact market severity rather than relying on heuristic scoring.

- Historical Backtesting Engine: Build a module to run our NLP signals against years of historical stock data to rigorously prove the algorithm's actual ROI and risk reduction metrics before deploying live capital.